Machine Learning vs. Deep Learning: Two Real-World Case Studies

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Machine learning (ML) and deep learning (DL) both help computers learn from data. They differ in how they find patterns. In traditional ML, people choose the features that matter. In DL, a neural network with many layers learns the features on its own. This report examines one example of each from the lesson. Customer churn prediction represents simple ML, and image recognition represents DL. For each, I describe a real-world application, explain why the approach fits, and explain why the other approach does not.

Example 1: Customer Churn Prediction (Simple Machine Learning)

Real-World Application

SyriaTel, a telecom company in Syria, built a system to predict which customers were likely to leave (Ahmad et al., 2019). The team used customer records such as call activity, billing, and service usage. They also added social network features that described who each customer communicated with. They tested several traditional ML algorithms, including decision trees, random forests, and gradient-boosted trees. The best model, XGBoost, reached an area under the curve (AUC) of 93.3%. The company could then target at-risk customers with retention offers before they left.

Why Simple Machine Learning Is Suitable

First, the data is structured. Customer records fit neatly into rows and columns, and each column has a clear meaning. This is the type of data traditional ML handles well.

Second, people already know which factors matter. Telecom analysts understand that contract length, monthly charges, and service complaints affect churn. They can design these features directly. Ahmad et al. (2019) found that adding well-designed features, such as the social network features, clearly improved the model’s performance.

Third, the results must be explained. A business needs to know why a customer is at risk, not just that they are. Tree-based models can show which features drove each prediction. Managers can then act on clear reasons, such as offering a better plan to customers with high bills.

Finally, traditional ML is efficient. It trains quickly on standard servers. The model can be retrained often as customer behavior changes.

Why Deep Learning Is Not Suitable

Deep learning would add cost without clear benefit here. Its main strength is learning features from raw, unstructured data, but churn data is already structured and meaningful. On tabular data like this, a deep network often performs no better than a well-tuned tree model. It also needs more data, more computing power, and more tuning. Most importantly, a deep network is harder to explain. If a model flags a customer but cannot show why, the business cannot design the right response or justify its decisions.

Example 2: Image Recognition with a CNN (Deep Learning)

Real-World Application

Diabetic retinopathy is an eye disease caused by diabetes that can lead to blindness if it is not caught early. Gulshan et al. (2016) at Google trained a convolutional neural network (CNN) on about 128,000 retinal photographs. Each image had been graded by eye doctors. The trained model detected referable diabetic retinopathy with accuracy comparable to that of the ophthalmologists. This type of screening can help clinics that have few eye specialists find patients who need treatment.

Why Deep Learning Is Suitable

First, the input is raw images. A retinal photo is millions of pixel values with no labeled columns. A CNN learns directly from these pixels. Its early layers detect simple patterns such as edges. Its deeper layers combine them into signs of disease, such as tiny bleeds and swelling.

Second, the important features are too complex to define by hand. Doctors recognize disease through years of visual experience. It is very hard to turn that experience into exact rules, such as the precise size, shape, and color of every lesion. Deep learning removes this barrier because it discovers the features on its own.

Third, the project had what deep learning needs. It had a very large set of labeled images and the computing power to train a deep network. Under these conditions, DL can reach expert-level performance.

Why Simple Machine Learning Is Not Suitable

A traditional ML model cannot work well with raw pixels. People would first need to measure features by hand, such as counting lesions or measuring blood vessels. This work is slow, depends on expert time, and still misses subtle patterns. The final model would only be as good as these hand-made features. In computer vision, this limitation is the main reason deep learning has largely replaced older feature-based methods.

Comparison

Table 1 summarizes how the problem itself determines the right approach.

Table 1

Simple ML vs. Deep Learning for the Two Case Studies

Factor	Customer Churn (Simple ML)	Retinal Images (Deep Learning)
Data type	Structured tables of customer records	Unstructured raw pixel images
Features	Defined by people (e.g., call usage, contract length)	Learned automatically by the network
Data and computing	Works with moderate data and standard servers	Needs very large labeled image sets and GPUs
Explainability	High; managers can see which factors drive churn	Low; harder to explain individual predictions
Best fit	Business decisions that must be explained and acted on	Visual pattern recognition at expert level

Conclusion

Neither approach is better in every case. The right choice depends on the type of data, how much data is available, and whether results must be explained. Customer churn prediction uses structured data with known factors, and its results drive business decisions, so simple ML is the better fit. Retinal image screening uses raw images with patterns too complex to define by hand, so deep learning is the better fit. As a software developer, I see this as a design decision

similar to choosing the right tool for a system: the simplest model that solves the problem well is often the best one.

References

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AI Use Summary

I used Claude (Anthropic) to help identify case studies, organize the report, and draft the text. I reviewed the content against the cited sources and edited it to reflect my own understanding.